

13. Construct a C program for implementation of the various memory allocation strategies.

AIM

To write a C program to implement various memory allocation strategies such as **First Fit, Best Fit and Worst Fit.**

CODE

```
#include <stdio.h>

int main()
{
    int block[10], process[10];
    int nb, np, choice;
    int i, j, index;

    printf("Enter number of blocks: ");
    scanf("%d", &nb);

    printf("Enter block sizes:\n");
    for(i = 0; i < nb; i++)
        scanf("%d", &block[i]);
    printf("Enter number of processes: ");
    scanf("%d", &np);
    printf("Enter process sizes:\n");
    for(i = 0; i < np; i++)
        scanf("%d", &process[i]);
    printf("\n1. First Fit");
    printf("\n2. Best Fit");
    printf("\n3. Worst Fit");
    printf("\nEnter choice: ");
    scanf("%d", &choice);
```

```

for(i = 0; i < np; i++)
{
    index = -1;

    for(j = 0; j < nb; j++)
    {
        if(block[j] >= process[i])
        {
            if(choice == 1)
            {
                index = j;
                break;
            }
            else if(choice == 2)
            {
                if(index == -1 || block[j] < block[index])
                    index = j;
            }
            else if(choice == 3)
            {
                if(index == -1 || block[j] > block[index])
                    index = j;
            }
        }
    }

    if(index != -1)
    {

```

```

        printf("Process %d allocated to Block %d\n",
               i + 1, index + 1);

        block[index] = block[index] - process[i];
    }
else
{
    printf("Process %d cannot be allocated\n", i + 1);
}
}

return 0;
}

```

OUTPUT:

main.c	Run	Output
<pre> 1 #include <stdio.h> 2 3 int main() 4 { 5 int block[10], process[10]; 6 int nb, np, choice; 7 int i, j, index; 8 9 printf("Enter number of blocks: "); 10 scanf("%d", &nb); 11 12 printf("Enter block sizes:\n"); 13 for(i = 0; i < nb; i++) 14 scanf("%d", &block[i]); 15 16 printf("Enter number of processes: "); 17 scanf("%d", &np); 18 19 printf("Enter process sizes:\n"); 20 for(i = 0; i < np; i++) 21 scanf("%d", &process[i]); 22 23 printf("\n1. First Fit"); 24 printf("\n2. Best Fit"); 25 printf("\n3. Worst Fit"); 26 printf("\nEnter choice: "); 27 scanf("%d", &choice); 28 29 for(i = 0; i < np; i++) 30 { 31 index = -1; 32 33 for(j = 0; j < nb; j++) 34 { 35 if(block[j] >= process[i]) 36 </pre>		<pre> Enter number of blocks: 3 Enter block sizes: 23 34 21 Enter number of processes: 2 Enter process sizes: 23 43 1. First Fit 2. Best Fit 3. Worst Fit Enter choice: 1 Process 1 allocated to Block 1 Process 2 cannot be allocated === Code Execution Successful === </pre>

RESULT

Thus, the C program to implement **First Fit, Best Fit and Worst Fit** memory allocation strategies was successfully executed.